

Custom Food Shop Application

Capstone 2

Version 7.1 Y

Setup

Disclaimer

This version of the capstone is more difficult. There are more steps that you will have to complete to finish this capstone. There are less guidelines on how to complete this capstone project. If you want to do this version of the capstone, please discuss with your instructor and get approval BEFORE YOU START.

Capstone Setup

For this capstone you should have a public GitHub repo. Make sure that the capstone repo has a meaningful name (do not give it a generic name).

Create the project repo in the `C:/pluralsight/capstones` folder.

Create a GitHub project board to manage your work. Use the project requirements to create user stories on your board so that you can manage your time throughout the project.

Be sure to read all project requirements before beginning to code.

Capstone Description

This project is the point of sales application, for your own custom shop. Currently your customers will be able to customize their order. Until recently we have been managing all orders in person and are currently taking all orders on paper. But our business is growing, and we need a way to automate the order process (and eventually even make it available online).

You will need to customize this shop, but still follow the requirements. You will need to choose what item your store sells, how it can be specialized, the types of that item, toppings, etc. You will need to choose a name for your business as well.

This application should take full advantage of Object Oriented Analysis and Design. In other words, you should make use of OOP concepts throughout this process.

Create classes and interfaces as necessary to build this solution. As you begin this project you should start by creating a class diagram of the classes, interfaces that you will need. This will be your starting point, and it may change as your application progresses throughout the week. You should keep your diagram current. You should also save the diagram to your repository.

HINT: Use the requirements below to look for nouns and verbs to help you create your class diagram.

Application Requirements

Pick one type of item that is core to your business, like a sandwich, taco, or pizza in the cases of the other stores. Customers must be able to order your item in 3 sizes.

When they order, they choose the type of that they would like. For a sandwich case, wheat, or white. The customer can then choose their toppings. Toppings are categorized as regular and premium. Proteins and cheeses are premium toppings, but most other toppings are considered regular.

Customers can request for extra toppings, but premium toppings come at an additional cost. Each item should also have a special option, like a pizza with stuffed crust.

A customer can place an order with 0 or more items on the order. If a customer places an order with 0 items, they must purchase a side or a drink.

When a customer places the order, they should be prompted to customize each item one at a time.

A customer should also be able to add drinks and a side to their order.

When they have completed their order, the application should display the order details, including the list of items that were ordered with all the toppings so that the customer can verify that the order is correct. The screen should also display the total cost of the order.

When the customer completes the order, the order details should be saved to a receipts folder. Each order should have its own receipt file, and it should be named by the date and time that the order was placed

(yyyyMMdd-hhmmss.txt - i.e. 20230329-121523.txt)

Item Prices	Size 1	Size 2	Size 3
Type - Type 1 - Type 2 - Type 3 - Type 4	3.50	9.00	8.50
Toppings	Size 1	Size 2	Size 3
Meats - Meat 1 - Meat 2 - Meat 3 - Meat 4 - Meat 5 - Meat 6	1.00	2.00	3.00
Extra Meat	.50	1.00	1.50
Premium Toppings - Premium 1 - Premium 2 - Premium 3 - Premium 4	.75	1.50	2.25
Extra Cheese	.30	.60	.90
Regular Toppings - Regular 1 - Regular 2 - Regular 3 - Regular 4 - Regular 5 - Regular 6 - Regular 7 - Regular 8 - Regular 9	Included	Included	included
Condiment - Sauce 1 - Sauce 2 - Sauce 3 - Sauce 4 - Sauce 5 - Sauce 6	Included	Included	Included
Sides - Side item 1 - Side item 2	Included	Included	Included

Other Products	Small	Medium	Large
Drinks	2.00	2.50	3.00
Main Side	1.50		

Your application must include several screens with the listed features in order to be considered complete:

- **Home Screen**
 - The home screen should give the user the following options. The application should continue to run until the user chooses to exit.
 - **1) New Order**
 - **0) Exit** - exit the application
- **Order Screen** - All entries should show the newest entries first
 - **1) Add Item**
 - **2) Add Drink**
 - **3) Add Main Side**
 - **4) Checkout**
 - **0) Cancel Order** - delete the order and go back to the home page
- **Add Item** - the add item screen will walk the user through several options to create the item
 - **Select your type:**
 - **Item size:**
 - **Toppings:** - the user should be able to add extras of each topping
 - **Meat:**
 - **Premium topping:**
 - **Other toppings:**
 - **Select condiments:**
 - **Would you like the item specialized?**
- **Add Drink** - select drink size and flavor
- **Add Main Side** - select main side type
- **Checkout** - display the order details and the price
 - **Confirm** - create the receipt file and go back to the home screen
 - **Cancel** - delete order and go back to the home screen

(Optional Bonus) Challenge Yourself

If you have time and want to challenge yourself, consider the following:

Create several Signature Items that are a template for what the customer can order. The customer could then customize the existing toppings to either remove or add more toppings to the item.

HINT: Signature Items could be managed by creating custom classes that inherit from the Item class.

Other General Project Requirements

Your project must also meet the following requirements:

Repository

- Your code must be in a public GitHub repository
- You must use Git branches appropriately
- The repository must contain an appropriate Git commit history
 - **Minimally**, you should have a commit for each meaningful piece of work completed
- It must contain an informative README file that:
 - Describes your project
- Include exports or screenshots of your diagrams

Class Demonstrations

Each student will be given 10 minutes to demonstrate their project to the class on "project demonstration day". During this time, you will:

- Present your application - run through the different screens and scenarios
- Describe / show one interesting piece of code that you wrote
- Answer questions from the audience if time permits